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Boundary-spanning search and business model innovation: the joint moderating effects of innovative cognitive imprinting and environmental dynamics

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Abstract

Purpose – Business model innovation (BMI) is an important channel of enterprise innovation, and BMI's antecedents have attracted extensive attention. The purpose of this paper is to address a substantial gap in the extant literature by developing a moderate model to explain the effects of boundary-spanning search on BMI as well as whether and how innovative cognitive imprinting (ICI) and environmental dynamics (ED) affect the above relationship.

Design/methodology/approach – A total of 239 usable questionnaires from different enterprises in China were collected to obtain firm-level data. Then multiple regression analyses were used by SPSS software to test hypotheses.

Findings – Boundary-spanning search extensity (BSE) and focus have inverted *U*-shaped impacts on BMI; ICI moderates the relationship between boundary-spanning search and BMI and steepens the curves; ED weakens the moderating role of ICI.

Originality/value – By identifying two antecedents of BMI, this paper contributes to the literature on the antecedents of BMI. Meanwhile, the joint moderating effect of ICI and ED is introduced into the emergent analysis framework of the relationship between boundary-spanning search and BMI and examined through empirical analysis for the first time.

Keywords Business model innovation, Boundary-spanning search, Innovative cognitive imprinting, Environmental dynamics, Search extensity, Search focus

Paper type Research paper

1. Introduction

Changes in the business environment and market competition, against the backdrop of digital economy, lead to challenges in competition rules and reshape the existing competition pattern (Chiara *et al.*, 2020; Penttilä *et al.*, 2020). In this context, compared with product innovation and process innovation, a new subject of innovation (Zott *et al.*, 2011), i.e. business model innovation (BMI), has become a new track for enterprises to seize the market highland and change the inherent competitive situation (Jin *et al.*, 2021). BMI has also been perceived as an important means to realize the value of enterprises' innovative behavior, such as technological innovation and product innovation, and it enables enterprises to obtain considerable economic returns while reducing uncertainty (Snihur *et al.*, 2018). After decades of development, BMI has become an important research field in strategic management and innovation literature (Foss and Saebi, 2017). Extant literature has revealed the strong relationships between BMI and enterprise performance (Rietveld, 2018; Sohl *et al.*, 2020; Zott and Amit, 2007), competitive advantage (Amit and Zott, 2012; Casadesus-Masanell and Zhu, 2013; Snihur *et al.*, 2018; Teece, 2018) and firm sustainability (Pedersen *et al.*, 2018).

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Consequently, it is not biased that business models have been one of the key channels of innovation.

BMI is defined as "designed, novel, nontrivial changes to the key elements of a firm's business model and/or the architecture linking these elements" (Foss and Saebi, 2017). While attracting extensive attention from practitioners and academia, research on the antecedents of BMI has become a hot issue (Foss and Saebi, 2017, 2018). Some recent research emphasizes the external antecedents of BMI, such as new technology (Wirtz *et al.*, 2010), cooperative networks (Bao *et al.*, 2020) and environmental factors (Pati *et al.*, 2018; Schmitt *et al.*, 2018), whereas others highlight internal antecedents, such as enterprise strategic orientation (Hock-Doepgen *et al.*, 2021; Klein *et al.*, 2021; McDonald and Eisenhardt, 2020), organizational capability (Miroshnychenko *et al.*, 2021; Soluk *et al.*, 2021), top managers (Aspara *et al.*, 2013; Narayan *et al.*, 2021) and organizational learning (Foss and Saebi, 2017; Sosna *et al.*, 2010; Yuan *et al.*, 2021). However, there is a lack of comprehensive research on the integration mechanism between internal active-controlled behavior and external environment and how it affects BMI in a comprehensive framework. The connecting attribute of the business model shows that it not only connects the upstream and downstream partnerships of the focal enterprise (Amit and Zott, 2015) but also connects the external environment of the enterprise (McDonald *et al.*, 2021; Peprah *et al.*, 2021). Furthermore, to compete effectively in the rapidly changing environment, a critical ability of enterprises is to obtain knowledge from the outside, i.e. the ability to mine new ideas from outside their boundaries is becoming increasingly essential (Katila and Ahuja, 2002; Laursen and Salter, 2006). Therefore, we suspect that enterprises' active exploratory learning behavior may have a non-negligible impact on BMI.

Knowledge-based innovation literature has established a strong correlation between knowledge and enterprise innovation (Kafouros *et al.*, 2008; Katila and Ahuja, 2002; Laursen and Salter, 2006; Lewin *et al.*, 2020; Rosenkopf and Nerkar, 2001). Furthermore, opportunities can be created through the activities of acquiring, learning and utilizing distinct knowledge (Alvarez *et al.*, 2013; Barreto, 2012; Mitchell *et al.*, 2008; Tippmann *et al.*, 2017). Organization learning is a process through which experience performing a task is converted into knowledge (Argote *et al.*, 2021). Therefore, this study defines organizational learning as the process of acquiring, absorbing and transforming knowledge in order to supplement knowledge in the changing market environment. As a key process of organizational learning, search is an effective means for organizations to acquire knowledge, and it is intertwined with knowledge creation and knowledge transfer (Argote *et al.*, 2021). Search is defined as a "controlled and proactive process of attending to, examining, and evaluating new knowledge and information" (Li *et al.*, 2013), which is an integral process of knowledge-based innovation theory (Carlile, 2004; Cohen and Levinthal, 1990; Grant, 1996; Snihur and Wiklund, 2019; Trantopoulos *et al.*, 2017). Enterprises need to actively or passively search for knowledge, obtain information and integrate resources in the external environment in order to achieve BMI. Existing literature also suggests that searching for external knowledge of enterprises may help to obtain general business knowledge so as to generate motivation and ideas for new business models (Berends *et al.*, 2016; Martins *et al.*, 2015; Rezazade Mehrizi and Lashkarbolouki, 2016).

There are several boundaries in the organization (Chesbrough *et al.*, 2018; Lopez-Vega *et al.*, 2016; Rosenkopf and Nerkar, 2001). Boundary-spanning search in this study is defined as the search activity across the organizational (or enterprise) boundary to achieve organizational goals in a dynamic environment. Previous research has established that breadth and depth are the two distinct dimensions of search (Katila and Ahuja, 2002; Laursen and Salter, 2006; Leiponen and Helfat, 2010). Meanwhile, combined with the relevant research on the diversity of knowledge sources and channels (Henttonen and Ritala, 2013), we identify boundary-spanning search extensity (hereafter referred to as BSE), widely acquiring knowledge from multiple sources or channels, and boundary-spanning search focus (hereafter referred to as BSF), deeply understanding knowledge in some specific fields. Further, the organization's time, ability, attention and resources are limited (Li *et al.*, 2013; Terjesen and Patel, 2017). It indicates that the knowledge sources of boundary-spanning search cannot be expanded or deepened indefinitely. Therefore, based on organizational learning theory, this study examines the impact curve of BSE and BSF on BMI from the impact of organizational learning on innovation.

The environment, including the internal and external environment, is an important context for organizational learning. It interacts with organizational experience and then affects organizational learning (Argote *et al.*, 2021; Argote and Miron-Spektor, 2011). BMI also means changes in the content and structure of existing business models (Chesbrough, 2010; Snihur and Zott, 2020; Zott and Amit, 2007, 2010), which implies that activities taken to innovate business models would be affected by the existing state and environmental changes in the organization. Meanwhile, the organizational imprinting theory suggests that the strategy, structure, culture and decision-making rules are embedded in the organization in sensitive periods and continue to exist while the environment changes (Marquis and Tilcsik, 2013; Simsek *et al.*, 2015; Sinha *et al.*, 2020). In particular, organizations affect the thinking, desire, learning and memory of their members through their sense of values, organizational culture and other forms (Snihur and Zott, 2020). Such an imprinting is called "cognitive imprinting." Obviously, search behavior aimed at innovating business models will be continuously and permanently affected by the members' innovative cognition, which is a new embodiment of the organization's historical experience. Meanwhile, the change in the environment will lead to challenges to the cognitive imprinting that has already existed in the organization (De Cuyper *et al.*, 2020), and then it will lead to a conflict between the original imprinting and the new environment. Therefore, it is necessary to investigate the mechanisms by which cognitive imprinting and environmental change influence the relationship between boundary-spanning search and BMI.

Summing up, based on organizational learning theory and organizational imprinting theory, this study brings the influence of enterprise active-controlled behavior and environmental change into a comprehensive research framework, constructs a theoretical model of BSE and BSF on BMI and introduces innovative cognitive imprinting (hereafter referred to as ICI) and environmental dynamics (hereafter referred to as ED) as moderating variables. To test the theoretical model, this study collected data from 239 enterprises by survey and implemented multiple regression analysis. This study aims to make several contributions in the following ways. First, considering the diversity of knowledge sources and channels, boundary-spanning search is divided into BSE and BSF. Then, the non-linear impacts of these two types of boundary-spanning search on BMI are analyzed, which expands the scope of boundary-spanning search literature as well as the research of antecedents of BMI. Second, new boundary conditions of enterprise BMI are expounded by revealing the joint moderating role of ICI and ED. While highlighting the systematic and deliberate intersection, integration and application of organizational learning theory and organizational imprinting theory in BMI, this study emphasizes enterprises' cognition of context and provides theoretical support for enterprises' boundary-spanning search actions and BMI.

2. Theoretical analysis and hypotheses

2.1 Boundary-spanning search and business model innovation

An innovative business model requires enterprises to actively seek and utilize external heterogeneous information, knowledge and resources in order to expand the existing innovation endowment (Lewin *et al.*, 2020). For knowledge demand, BMI differs from traditional technological innovation in that it relies more on general business knowledge rather than specific technical knowledge, and it is closely related to boundary-spanning search behavior (Snihur and Wiklund, 2019). Innovating business models requires a tight understanding of the existing business models in the market, obtaining information timely, as well as finding and making full use of opportunities. Boundary-spanning search has gone beyond just a knowledge-searching behavior and has become a critical tactic for enterprises to access information, identify opportunities and expand innovative resources. Enterprises need to obtain heterogeneous resources in a dynamic environment, capture external policy information and maintain communication with corresponding government departments (Barron and Pereda, 2020), so as to correctly predict the imminent market changes (Hsu and Lim, 2014). Therefore, searching for heterogeneous knowledge outside the organizational boundaries is essential for BMI.

The term BSE refers to the extensity degree to which enterprises use different strands of channels or sources to gain information, resources and other knowledge. More exploration is needed when introducing new business models that rely on non-technical knowledge or tacit business knowledge (Snihur and Wiklund, 2019). Furthermore, innovation needs vast creative sources. A broader view can help enterprises obtain a multitude of heterogeneous knowledge resources to improve their success rate of innovation. Meanwhile, a business model involves all participants in value-adding activities, including consumers, suppliers and focal enterprises. This also leads to higher requirements for focal enterprises to capture external information, resources and other knowledge. Increased extensity means a wider range of knowledge sources, which can increase the differentiation and diversity of available knowledge (Katila and Ahuja, 2002). Thus, innovation performance can be improved by creating new combinations of knowledge (Rosenkopf and Nerkar, 2001). At the same time, extensive boundary-spanning search can enable enterprises to obtain the market news and competition status of their own industry and other industries, which is conducive to expanding horizons, locating their own positioning, finding market opportunities and putting forward new value propositions for value creation. In general, extensive search helps to obtain innovative inspiration, learn from the instructive factors and increase the innovativeness of the business model. However, due to the constriction of "absorptive capacity" (Lewin *et al.*, 2020), "timing" (Katila and Chen, 2008) and "attention distribution" (Ocasio, 1997), the problem of excessive search may also occur (Laursen and Salter, 2006). Specifically, first, the influx of excessive external knowledge may exceed the existing information processing capacity of the enterprise, resulting in resource congestion. Second, due to the dynamics of the environment, innovative ideas may come untimely. That means BMI cannot match the times and that would eventually lead to failure. Finally, the pressure of coordinating multiple knowledge resources may negatively affect innovation performance, as paying too much attention to external knowledge will lead to the neglect of internal innovation resources. Meanwhile, the BSE means exploring many unfamiliar fields. However, searching in unfamiliar fields is more challenging and expensive (Li *et al.*, 2013; Terjesen and Patel, 2017). Therefore, we expect that excessive BSE will have a negative impact on BMI.

H1. There is an inverted *U*-shaped relationship between BSE and BMI.

The term BSF refers to the concentration degree of obtaining knowledge such as information and resources from different external channels or sources. BSF also means that enterprises establish in-depth and long-term cooperative relations with some specific knowledge sources. Finding new ideas also requires maintaining frequent interaction with some knowledge sources (Henttonen and Ritala, 2013). It enables enterprises to capture knowledge pertaining to the entrepreneurial experience, operation strategy and resource network of the targeted industry. That would help them to improve their processes, reorganize their business model elements and then achieve BMI by expanding their innovation network and integrating knowledge and resources. Deep search in some specific fields is also conducive to the reliability of search (Levinthal and March, 1981). Focusing on specific fields makes the targeted domain more familiar. The profound exploration of knowledge can positively impact the reduction of the occurrence of errors and failures (Henttonen and Ritala, 2013). Meanwhile, increased search focus is conducive to improving the ability to identify valuable knowledge (Katila and Ahuja, 2002), and it would increase the possibility of successfully promoting innovation through the comprehension of knowledge. For example, for product innovation, searching in a narrow range can lead to better innovation performance (Albert and Siggelkow, 2021). However, excessive in-depth searching will limit the maximum value of search results (Hsu and Lim, 2014). Continuous and in-depth searching in a certain field may yield core rigidity (Leonard, 1995). That means excessive BSF would negatively affect the innovation of a business model. While searching, the active subject cannot predict which kinds of knowledge and resources are valuable or which have the highest return rate (Laursen and Salter, 2006). Excessive investment in some useless content would cause a waste of organizational resources. Excessive focus also implies heavy dependence on some specific resources or partners. For example, it strongly depends on a few external resources that provide essential knowledge input (Snihur and Wiklund, 2019). This kind of resource dependence will cause obstruction to BMI. Meanwhile, existing studies also suggest that relying heavily on a few types of knowledge sources may limit the ability of enterprises to create innovation (Chiang and Hung, 2010). Therefore, we expect that excessive BSF would exert a negative impact on BMI.

H2. There is an inverted *U*-shaped relationship between BSF and BMI.

2.2 The moderating role of innovative cognitive imprinting

According to the organizational imprinting theory (Johnson, 2007; Marquis and Tilcsik, 2013; Stinchcombe, 2000), the early conditions of an organization's establishment will persist and affect the later results of the organization. Organizational imprinting mainly acts on three areas as follows: organizational culture and social ties, organizational memory and identity and organizational collective motivation and goals (Bluck *et al.*, 2005; Bryant, 2014). Specific to cognitive imprinting, the thinking, desire, learning and memory of organization members will be affected by cognitive imprinting (Snihur and Zott, 2020). Meanwhile, the impact of cognitive imprinting on the cognition of organization members will persist and spread widely (Ellis *et al.*, 2017).

For the function of cognitive imprinting, it can be inferred that ICI affects the innovative thinking, innovative desire, innovative learning and innovative memory of organization members. With the persistence of ICI, organization members could internalize their cognition of innovation into their own consciousness. This sense of innovation urges members to actively pay attention to innovative behavior and explore experiential learning and innovative resources behind innovation. Specifically, the role of ICI is reflected in the following three aspects: First, ICI helps to match the innovative behaviors among organizational members. Organizations can leave innovative cognitive marks on their members through founders (Ellis *et al.*, 2017; Marquis and Tilcsik, 2013), top management teams (Snihur and Zott, 2020), etc. In a new learning environment, search behavior is related to managers' cognition of new problems and their ways of making strategies based on this cognition (Tripsas and Gavetti, 2000). The special status of founders or top managers makes their boundary-spanning search behavior widely accepted or even imitated by other organization members. Therefore, the strong ICI makes the effect of boundary-spanning search on BMI more significant; that is, it deepens the impact of boundary-spanning search on BMI. Second, ICI is conducive to the matching between organizational members' innovative search behaviors and organizational goals.

Organizational members can be attracted to organizational imprinting and its potential value (De Cuyper *et al.*, 2020). Therefore, ICI can affect members' understanding of the organizational vision and objectives of BMI. Under the influence of an innovative culture, atmosphere and values, it is not hard for organization members to form innovative thinking and then look for opportunities outside organizational boundaries that may facilitate BMI. When the ICI is strong, members have a stronger sense of identity with the organizational vision and objectives, which drives the members to actively adapt to and promote the business model change that is in line with the organization's objectives. Meanwhile, innovative search behaviors in line with organizational collective motivation and goal expectations would be recognized and encouraged. That means the boundary-spanning search behaviors aimed at innovating business models are easier to implement when the imprinting of innovative cognition is stronger. Finally, ICI is conducive to the coordination of organizational innovative resources. BMI involves the acquisition, binding and utilization of internal and external resources of the organization. Under the influence of ICI, organization members' resource coordinating activities aimed at innovating business models would be supported by other members. In summary, a strong ICI would enable organizations to coordinate their resources and obtain the support of stakeholders more effectively.

As BSE increases from low to medium level, strong ICI enables the organization to deal with resources efficiently according to the existing cognitive routine to promote the transformation and utilization of resources. However, when BSE exceeds a certain threshold, excessive heterogeneous resources will flow into the organization. This brings challenges to the organization's resource processing ability. Meanwhile, the existing innovative cognition cannot be updated in time, and some resources that may be conducive to BMI will be ignored because of the falloff in innovative cognition, which aggravates the negative impact of BSE on BMI. For BSF, with the expansion of focus from low to medium level, resource sources are gradually concentrated. The cognitive inertia brought by strong ICI would also bring higher resource processing efficiency so as to promote the innovation of business models. However, as the focus of boundary-spanning search exceeds a certain threshold, the concentration of resource sources is further deepened, and the heterogeneity of resources is tough to guarantee. Meanwhile, the inertia brought by imprinting makes it difficult for organizations to shift their existing cognition in time, which intensifies the negative effect of BSF on BMI. Therefore, this study suggests that ICI would enhance the inverted *U*-shaped relationship between BSE, BSF and BMI:

H3a. ICI makes the inverted *U*-shaped relationship curve between BSE and BMI steeper.

H3b. ICI makes the inverted *U*-shaped relationship curve between BSF and BMI steeper.

2.3 The joint moderating role of ICI and ED

ED is defined as the frequency and amplitude of change in the external environment (Dess and Beard, 1984). The changes in information, resources and other knowledge in the environment affect the operation and innovation of business models. In this dynamic environment, external information, knowledge and social needs change rapidly, and enterprises may lose their leading position due to a lack of timely reaction. If reacted to timely and reasonably, these contents can act as important resources that promote BMI. However, ED also bring more threats to enterprises (Zhang and Zhu, 2021). Enterprise behavior lags behind environmental change, and a highly changing environment is more likely to exert a negative impact on enterprise operations (Bao *et al.*, 2020). At the same time, the outcome of boundary-spanning search will be negatively affected by the dynamics. In a highly changing environment, existing products and services are liable to become obsolete (Jansen *et al.*, 2005).

In other words, the fast change of information, resources and other knowledge would make the searched content lose its value quickly.

ED would affect the moderating effect of the existing ICI. Imprinting theory indicates that the cognitive imprinting may expand, enhance or decline (De Cuyper *et al.*, 2020; Sinha *et al.*, 2020). When the environment changes, affected by internal and external factors, the imprinting will re-imprint in the sensitive period after creation (De Cuyper *et al.*, 2020; Marquis and Tilcsik, 2013; Muñoz *et al.*, 2018). This means that the existing ICI could be modified, updated or weakened if environmental changes occur. Formatted in the past period, ICI may not be able to adapt to the current context after suffering drastic environmental changes. That means existing cognitive structures may not match the innovation objectives of the organization. High ED means rapid changes in information, resources, products and services. In this situation, the innovation of business model faces a high risk of failure (Bao *et al.*, 2020). Considering the increased risk of failure of BMI and the negative impact on the acquisition of innovative knowledge, this study suggests that the increase in ED weakens the moderating effect of the original ICI.

H4a. High ED weakens the moderating effect of ICI on BSE and BMI.

H4b. High ED weakens the moderating effect of ICI on BSF and BMI.

The research model is shown in Figure 1.

3. Method

3.1 Samples and data collection

To test hypotheses, we collected data through a survey in China. Our questionnaire was primarily formed by the mature and widely used scales developed in an English context. The questionnaire was translated through double and reverse translation procedures. Subsequently, three researchers from related fields were invited to review and discuss the contents of the questionnaire and modify it according to the current situation of this study. After that, we invited three enterprise managers with enough working experience to give their opinions, and then we modified some contents to ensure the availability, rationality and feasibility of the data. Finally, in order to ensure a valid fit between the exercise enterprise and the content of the questionnaire, we conducted a pilot survey. A pilot survey was conducted in the Changyang Chuanggu national innovation and entrepreneurship campus, which is one of the four innovation and entrepreneurship blocks in Yangpu District, Shanghai, by means of field questionnaires. The enterprise directory was provided by the management department of the campus. A total of five enterprises were randomly selected. After getting the permission of the enterprises' managers, three respondents were selected from each enterprise for investigation through face-to-face interviews, and they completed the questionnaire on site. Based on the results and the feedback from the pilot survey, three

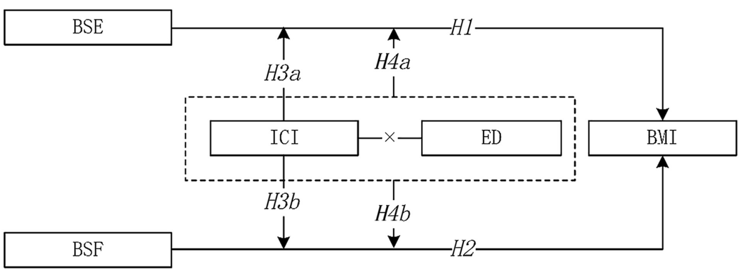


Figure 1.
Research model

researchers adjusted the content after full discussion, making the questions easier to understand and ensuring that respondents could fully understand the concept of items and the present situation of the innovation practice of enterprises' business models. Subsequently, we conducted a larger scale questionnaire distribution. Enterprises in Beijing, Shandong, Shanghai, Jiangsu, Guangdong and Shaanxi were selected as the research objects, covering the fields of intelligent manufacturing, information technology, enterprise service industry and medical health.

This study conducted the survey from March 2021 to April 2021 by adopting a combination of field research and online questionnaires. Due to resource constraints, this study used a convenient sampling method to collect data; this method is widely used in management and innovation studies (Wu *et al.*, 2019). Specifically, samples were gathered from the following sources: (1) randomly selecting enterprises in innovation campus in Shanghai (including Zhangjiang High-tech Industrial Development Zone, Changyang Innovation Campus, and others) and distributing questionnaires on site. A total of 65 questionnaires were distributed and 36 were collected. Subsequently, we conducted research in the Jinan High-tech Zone, Qingdao International Innovation Park and Xi'an High-tech Development Zone; obtained the list of enterprises with the help of the local management department and randomly selected enterprises for on-site distribution. In total, 97 questionnaires were distributed and 51 collected. (2) The university where the researcher is affiliated has a vast number of MBA/EMBA students from East China, South China and other regions in China. The industry of the students covers many fields, which provides a rich research sample for this study. Therefore, the study randomly selected MBA and EMBA students who were in school and graduated as respondents and sent 301 questionnaires via email, of which 167 were returned. (3) Through the researcher's social relations, we contacted enterprise employees in Beijing, Guangzhou, Shenzhen and other places. In total, 80 questionnaires were sent through e-mail or instant messaging software and 42 were completed. Because the research issues are at firm level, in order to ensure the quality of the survey, the respondents are mainly managers and technical staff. A total of 538 questionnaires were distributed through the field and online (including e-mail and communication software) distribution and 296 questionnaires were gathered. After drawing 57 questionnaires that did not meet the research requirements, where there were incomplete filling, inconsistent roles, illegible filling contents and incorrect filling of common sense questions, a total of 239 valid questionnaires were obtained, with an effective rate of almost 44%. We compared the enterprise age and scale of the samples received in two ways through the *t*-test. There was no significant difference in age and scale between on-site distribution and online distribution. Data from different sources can be combined for analysis.

Sample characteristics are shown in Table 1.

3.2 Measures

Our scales were based on existing scales, which were validated, and the necessary adjustments were made by considering the actual situation. Except for control variables, variables were measured using a seven-point Likert scale, where 1 = strongly disagree and 7 = strongly agree. Respondents were asked to indicate to what extent they agreed with the statements.

BMI: Based on the research of Zott and Amit (2007) and Guo *et al.* (2016), nine items were used to measure BMI and the Cronbach's α was 0.973.

BSE and BSF: Based on the research of Katila and Ahuja (2002), Henttonen and Ritala (2013) and Wang *et al.* (2020), BSE and BSF were each measured by five items, and their Cronbach's α were 0.957 and 0.964, respectively. Respondents were asked to what extent they agree with the items concerning obtaining knowledge from outside.

ICI: The ICI was measured in this study based on organizational members' perceptions of the innovation environment. Based on the scales of West (1990) and modified according to the

Characteristics	Range	Frequency	Percentage (%)
Industry	Manufacturing	54	22.59
	Traditional service	57	23.85
	Information technology	38	15.90
	Enterprise service	42	17.57
	Others	48	20.08
Firm age (years)	<1	4	1.67
	1–5	19	7.95
	6–10	39	16.32
	11–15	39	16.32
	>15	138	57.74
Firm scale (employees)	<100	42	17.57
	100–200	36	15.06
	201–500	39	16.32
	501–1,000	39	16.32
	>1,000	83	34.73
Ownership	State-owned	73	30.54
	Private-owned	72	30.13
	Mix-owned	60	25.10
	Foreign-owned	32	13.39
	Others	2	0.84

Table 1. Sample characteristics ($N = 239$)

research of Snihur and Zott (2020), seven items were used to measure the cognitive imprinting of innovation and the Cronbach's α was 0.963.

ED: Based on the research of Jansen *et al.* (2006), Schilke (2014) and Dubey *et al.* (2021), five items were used to measure ED and the Cronbach's α was 0.911.

Control variables: Referring to the research of Guo *et al.* (2016), we used four widely accepted control variables that may have impacts on enterprise BMI, including industry type, enterprise age, enterprise scale and property. Our samples include five industry types (i.e. manufacturing, traditional service, information technology, enterprise service and others). Firm age was measured by the years of establishment, ranging from less than 1 to more than 15. Firm scale was measured by the numbers of employees, ranging from less than 100 to more than 1,000. We controlled for ownership as five ownerships were presented in our samples (i.e. state-owned, private-owned, mix-owned, foreign-owned and others). All control variables were coded as 1 to 5 based on their ranges.

The detailed items of variables are shown in Table 2.

3.3 Common method variance

Necessary procedures to reduce the possible impact of common method variance (CMV) on the analysis have been adopted. However, each questionnaire was finished by only one respondent, and responses were subjective as well as susceptible to recall bias. Therefore, it was necessary to test for CMV. First, Harman's single-factor test was used to test whether there is a CMV. All latent variables were forced to load onto one factor using a confirmatory factor analysis (CFA). The result shows that the extracted single factor explains 45.3% of the total variance and does not reach 50% (Podsakoff *et al.*, 2003). Further, through CFA, it is found that the adaptation index of a one-factor model is worse than that of a five-factor model. Therefore, it can be considered that there is no serious CMV that would influence the study's finding.

3.4 Validity and reliability

All our scales were adapted from previous scales, which ensure their content validity. Cronbach's αs of variables were calculated by SPSS. As shown in Table 2, all Cronbach's α

Variables	Items		Factor loading
BMI (Cronbach's α = 0.973; CR = 0.977; AVE = 0.825)	(1)	Our business model offers new combinations of products, services and information	0.891
	(2)	Our business model attracts a lot of new suppliers and partners	0.909
	(3)	Our business model attracts a lot of new customers	0.917
	(4)	Our business model bonds participants together in new ways	0.921
	(5)	Our business model adopts new ways to transact	0.926
	(6)	We frequently introduce new ideas and innovations into our business model	0.903
	(7)	We frequently introduce new operational processes, routines, and norms into our business model	0.914
	(8)	We are pioneers of the business model	0.903
	(9)	Overall, our business model is novel	0.891
BSE (Cronbach's α = 0.957; CR = 0.967; AVE = 0.853)	(1)	We can widely search for information in unknown field	0.929
	(2)	We can widely search for knowledge about the market, policy, technology, etc.	0.922
	(3)	We devote the majority of our energy to creating diverse sources of knowledge	0.901
	(4)	We have many channels supplying resources conducive to innovation	0.934
	(5)	We focus on developing new external knowledge sources	0.931
BSF (Cronbach's α = 0.964; CR = 0.972; AVE = 0.874)	(1)	We concentrate on using information from familiar sources	0.943
	(2)	We have established in-depth cooperative relations with partners providing resources	0.929
	(3)	A small number of channels provide us with most of the information and resources	0.913
	(4)	We allocate much vigor to building a common understanding with our partners	0.935
	(5)	We conduct an in-depth search and obtain knowledge of an existing field	0.954
ICI (Cronbach's α = 0.963; CR = 0.969; AVE = 0.819)	(1)	Our firm recognizes the importance of innovation	0.905
	(2)	Our firm encourages employees to make bold innovations	0.919
	(3)	Employees will learn and practice the leader's innovative behavior	0.912
	(4)	Innovation is the goal and vision of our firm	0.906
	(5)	Innovation behavior will get a positive response from the company	0.882
	(6)	Communication between colleagues can bring new ideas	0.897
	(7)	Innovation is an important indicator when evaluating employee performance	0.914
ED (Cronbach's α = 0.911; CR = 0.934; AVE = 0.738)	(1)	The modes of production and service change often and in a major way	0.862
	(2)	The environmental demands on us are constantly changing	0.838
	(3)	Marketing practices in our industry are constantly changing	0.873
	(4)	Environmental changes in our industry are unpredictable	0.872
	(5)	In our environment, new business models evolve frequently	0.850

Table 2. Combined reliability and validity

values and combined reliability (CR) values of five variables are above 0.9, greater than the standard of 0.7, and the loading factors are greater than 0.5 (Nunnally, 1978). The above results indicate that the reliability of all constructs meet the requirements. Exploratory factor analysis (EFA) of variables shows that KMO value reaches 0.957 and the Bartlett test passes (p < 0.001), indicating that factor analysis is appropriate. The factor with a characteristic value greater than 1 is extracted by the maximum variance method, and all variables' factor loadings are greater than 0.8, indicating that there is high convergent validity.

We further tested the convergent validity and discriminant validity by CFA. In CFA, all items were loaded onto their corresponding potential structures, indicating good convergent validity (Anderson and Gerbing, 1988). At the same time, Table 3 also shows that the five-factor model has better adaptability than single-factor, two-factor, three-factor and four-factor models, which also indicates that the discriminant validity meets the requirements. All average variance extracted (AVE) values of potential variables are greater than 0.7, higher than the recommendation standard of 0.5 (Hair *et al.*, 2011) and greater than the correlation coefficient between variables (Fornell and Larcker, 1981), showing that the convergent validity and discriminant validity are satisfactory.

4. Results

4.1 Description and correlation

Statistical analysis was performed using SPSS software. Means, standard deviations and inter-scale correlations for variables are shown in Table 4. BSE and BSF are related to BMI, ICI and ED. Meanwhile, ICI is related to ED and BMI.

4.2 Main effects

Hierarchical multiple regressions were used to test the proposed relationships. According to the research of Haans *et al.* (2016), the inverted "*U*" effect should meet the following three conditions. (1) The coefficient of the quadratic term is significantly negative. (2) When the independent variable takes the lowest value, the curve slope is positive and significant, and when the independent variable takes the highest value, the curve slope is negative and significant. (3) The turning point of the curve is within the value range of the independent variable.

In terms of the effect of BSE on BMI, Model 2 in Table 5 shows that the coefficient of quadratic term (BSE2) is negative and significant ($\beta = -0.168$, $p < 0.01$). The value range of BSE after centralization is [−2.93, 3.07]. The slope when taking the lowest value is $k_{SBL} = 1.214$, and the slope when taking the highest value is $k_{SBH} = -0.802$. The value at the turning point of the curve is 0.685, falling within the value range after centralization. Its 95% confidence interval in untransformed values is [4.220, 5.761]. Therefore, BSE has an inverted *U*-shaped impact on BMI, and hypothesis H1 is supported.

In terms of the effect of BSF on BMI, Model 2 in Table 5 shows that the coefficient of quadratic term (BSF2) is negative and significant ($\beta = -0.279$, $p < 0.01$). The value range of BSF after centralization is [−3.02, 2.98]. The slope when BSF takes the lowest value is $k_{SDL} = 1.748$, the slope when BSF takes the highest value is $k_{SDH} = -1.600$. The value at the turning point of the curve is 0.113, falling within the value range after centralization. Its 95% confidence interval in untransformed values is [3.994, 4.395]. Therefore, BSF has an inverted *U*-shaped impact on BMI, and hypothesis H2 is supported.

Factor model	Variables	CMIN/df	GFI	NFI	CFI	RMSEA
One-factor model	BSE + BSF + ICI + ED + BMI	12.179	0.250	0.387	0.406	0.217
Two-factor model	BSE + BSF + ICI + ED, BMI	8.843	0.402	0.573	0.601	0.182
Three-factor model	BSE + BSF, ICI + ED, BMI	7.656	0.418	0.615	0.647	0.167
Four-factor model	BSE + BSF, ICI, ED, BMI	6.139	0.360	0.690	0.726	0.147
Four-factor model	BSE, BSF, ICI + ED, BMI	3.562	0.641	0.822	0.865	0.104
Five-factor model	BSE, BSF, ICI, ED, BMI	1.720	0.843	0.919	0.964	0.055

Table 3. Convergence validity and discriminant validity

4.3 Moderating effects

4.3.1 Moderating effects of innovative cognitive imprinting. According to the research of Haans *et al.* (2016), the moderating effect of an inverted *U*-shaped relationship is divided into two types: one is the change of curve turning point, and the other is the change of curve shape.

For BSE, with the increase of ICI, the turning point of the curve moves to the right and the highest point of the curve increases. Model 4 in Table 5 reveals a negative interactive coefficient which represents the joint effect of ICI and the quadratic of BSE (ICI*BSE2) regressed on BMI ($\beta = -0.041$, $p < 0.05$). When BSE is the minimum, the slopes of the curve under the low ICI and the high ICI are 0.378 and 0.968, respectively, both of which are positive. When the BSE is at its maximum, the curve slopes of high ICI and low ICI are −0.174 and −0.568, respectively. This means that ICI can make the curve steeper. When ICI increases, the turning point of the curve moves to the right. Therefore, H3a can be supported. The result is depicted in Figure 2(a).

For BSF, with the improvement of ICI, the turning point of the curve moves to the right and the highest point of the curve increases. Model 6 in Table 5 reveals a negative interactive coefficient which represents the joint effect of ICI and the quadratic of BSF (ICI*BSF2) regressed on BMI ($\beta = -0.037$, $p < 0.05$). When the BSF is at its minimum, the slope of the curve under the low ICI and the high ICI is 1.352 and 1.901, respectively, both of which are positive. When BSF is at its maximum, the curve slopes of high ICI and low ICI are −1.324 and −1.663, respectively, which are both negative. This means that ICI can make the curve steeper. When the ICI increases, the turning point of the curve moves to the right. Therefore, H3b is supported. The result is depicted in Figure 2(b).

The interactive effects at high versus low levels (Mean+1SD versus Mean−1SD) are plotted by the procedure illustrated by Toothaker (1994), and they are depicted in Figure 2.

4.3.2 Joint moderating effect of innovative cognitive imprinting and environmental dynamics. Model 8 in Table 5 shows a positive interactive coefficient representing the joint effect of ICI, ED, and the quadratic of BSE (ICI*ED*BSE2) regressed on BMI ($\beta = 0.055$, $p < 0.01$). H4a is supported. Meanwhile, Model 9 in Table 5 shows a positive interactive coefficient representing the joint effect of ICI, ED and the quadratic of BSF (ICI*ED*BSF2) regressed on BMI ($\beta = 0.021$, $p < 0.1$). Hypothesis H4b is supported. The joint interactive effects at high versus low levels (Mean+1SD versus Mean−1SD) are plotted and shown in Figure 3.

With the exception of low ICI and low ED, increasing ED causes the curve to steepen and the peak value to rise. This result is depicted in Figure 3(a). This means that high ED weakens the moderating role of ICI in the relationship between BSE and BMI. With a high ICI, the increase in ED weakens the effect that BSE has on BMI. With a low ICI, the increase in ED enhances the effect of BSE on BMI. When the ICI and ED are both at a low level, the quadratic term coefficient of the impact of BSE is pretty small ($\beta = -0.003$), indicating a nearly linear relationship. Meanwhile, *ceteris paribus*, the increase in ED makes the curve steep and the peak value increase. This means high ED weakens the moderating role of ICI in the relationship between BSF and BMI. This result is depicted in Figure 3(b).

5. Discussion

5.1 Main findings

Based on organization learning theory and organization imprinting theory, this study constructs a theoretical model covering BSE and focus, ICI, ED and BMI. By conducting an empirical test based on the survey data, this study has the following findings:

First, BSE and BSF have inverted *U*-shaped relationships with BMI, respectively. This indicates that there are thresholds for the effects of the two dimensions of boundary-spanning search on BMI, and BMI will be inhibited when extensity and focus increase to a pretty high

Table 4.
Means, stand deviation and correlations

Variables	1	2	3	4	5	6	7	8	9
1. Industry	1.000								
2. Firm age	0.063	1.000							
3. Firm scale	0.167***	0.109	1.000						
4. Ownership	−0.073	0.023	−0.100	1.000					
5. BMI	−0.182***	0.035	0.018	0.058	*0.908*				
6. BSE	−0.107	0.234***	0.179***	−0.009	0.457***	*0.924*			
7. BSF	−0.202***	0.059	0.122	0.031	0.412***	0.307***	*0.935*		
8. ICI	−0.085	0.080	0.137**	−0.021	0.524***	0.538***	0.307***	*0.905*	
9. ED	0.088	−0.088	0.065	−0.030	−0.583***	−0.211***	−0.282***	−0.225***	*0.860*
Means	N/A	N/A	N/A	N/A	3.443	3.930	4.023	4.443	4.954
S.D.	N/A	N/A	N/A	N/A	1.760	1.825	1.856	1.550	1.253
Minimum	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Maximum	5.000	5.000	5.000	5.000	6.889	7.000	7.000	6.857	7.000

Note(s): $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.001$; the diagonal italic numbers are the square root of AVE; N/A means not applicable for this item

Table 5. Regression results of hierarchical model

		Dependent variable: BMI								
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Constant		3.457^{***}	5.841^{***}	5.317^{***}	5.094^{***}	4.830^{***}	5.059^{***}	4.948^{***}	4.786^{***}	4.702^{***}
Control variables	Industry	-0.229^{***}	−0.081	−0.075	−0.070	-0.091^{**}	−0.062	-0.082^{*}	-0.104^{**}	-0.100^{**}
	Firm age	0.066	−0.110	-0.111^{*}	−0.086	−0.066	−0.084	−0.086	−0.065	−0.062
	Firm scale	0.058	−0.060	−0.034	−0.017	−0.017	−0.012	−0.016	−0.003	0.016
	Ownership	0.059	0.004	0.015	−0.001	0.029	−0.003	0.027	0.037	0.017
Independent variables	BSE		0.230^{***}	0.139^{***}	0.163^{***}	0.208^{***}	0.119^{***}	0.140^{***}	0.242^{***}	0.137^{***}
	BSF		0.063	0.035	0.005	0.050	0.056	0.092^{**}	0.017	0.120^{***}
	BSE2		-0.168^{***}	-0.085^{**}	-0.087^{**}	-0.089^{**}	-0.075^{*}	-0.066^{*}	-0.079^{**}	-0.069^{*}
	BSF2		-0.279^{***}	-0.245^{***}	-0.258^{***}	-0.216^{***}	-0.260^{***}	-0.250^{***}	-0.219^{***}	-0.243^{***}
Moderate variables	ICI			0.214^{***}	0.403^{***}	0.224^{***}	0.388^{***}	0.222^{***}	0.318^{***}	0.274^{***}
	ED			-0.335^{***}	-0.351^{***}	-0.557^{***}	-0.343^{***}	-0.549^{***}	-0.543^{***}	-0.541^{***}
Interactions	ICI*ED								-0.139^{**}	-0.175^{***}
	BSE*ICI				0.055^{*}				−0.021	
	BSE2*ICI				-0.041^{**}				-0.043^{*}	
	BSE*ED					-0.127^{***}			-0.121^{***}	
	BSE2*ED					0.058^{***}			0.022	
	BSF*ICI						0.051^{**}			0.054^{**}
	BSF2*ICI						-0.037^{**}			−0.017
	BSF*ED							−0.045		−0.052
	BSF2*ED							0.069^{***}		0.067^{***}
	BSE*ICI*ED								0.104^{***}	
	BSE2*ICI*ED								0.055^{***}	
	BSF*ICI*ED									−0.017
	BSF2*ICI*ED									0.021^{*}
	R^2	0.040	0.637	0.697	0.637	0.697	0.712	0.697	0.712	0.727
	Adjusted-R^2	0.023	0.625	0.684	0.625	0.684	0.697	0.684	0.697	0.713
	R^2 change	0.040^{***}	0.268^{***}	0.060^{***}	0.015^{***}	0.030^{***}	0.018^{***}	0.021^{***}	0.240^{***}	0.019^{**}
	F value	2.416^{*}	50.533^{***}	52.460^{***}	46.570^{***}	50.273^{***}	47.179^{***}	47.979^{***}	40.270^{***}	39.116^{***}

Note(s): $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.001$; BSE2 is the quadratic of BSE; BSF2 is the quadratic of BSF

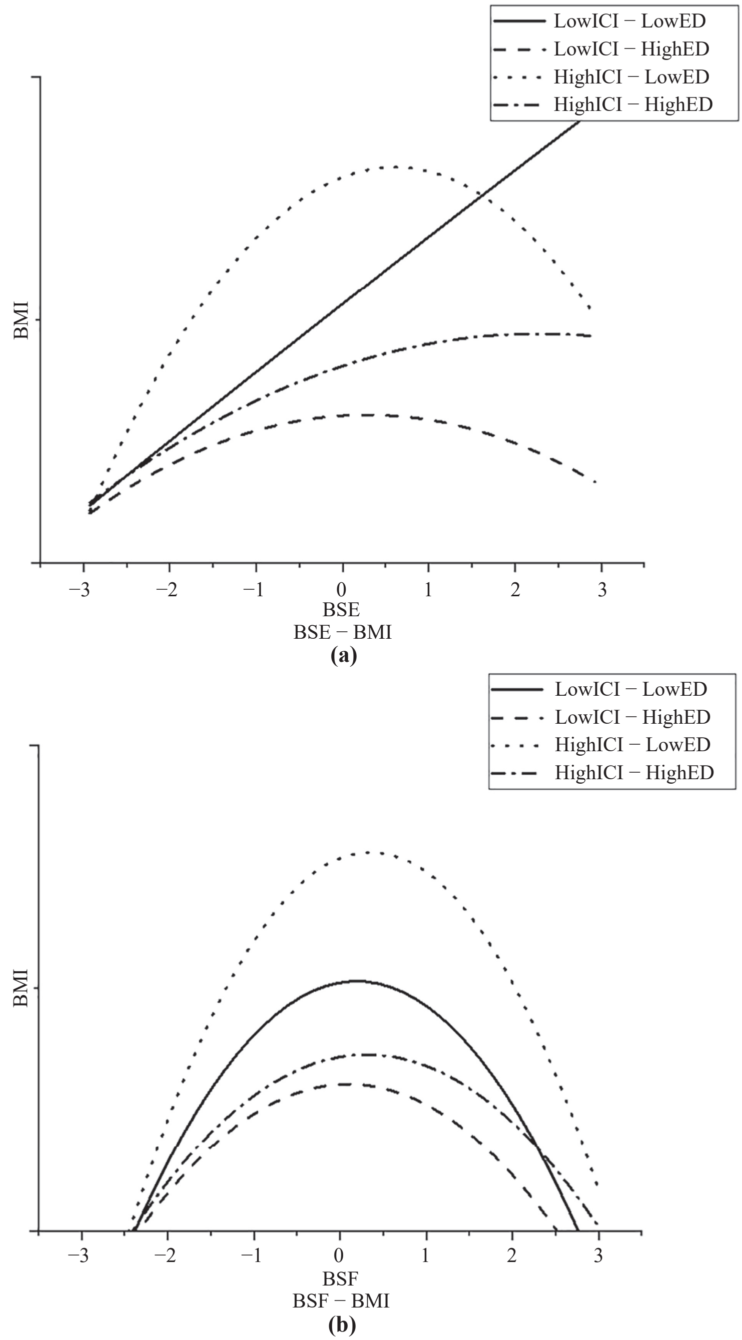


Figure 2.
Moderating effect

level, exceeding the thresholds. This is most likely due to the allocation of organizational resources, vigor and attention (McDonald and Eisenhardt, 2020) as well as absorptive ability (Argote *et al.*, 2021; Cohen and Levinthal, 1990), which would limit the positive role of external knowledge on innovation. For the antecedents of BMI, this study expands the finding of Snihur and Wiklund (2019) that boundary-spanning search negatively influences BMI, and a deeper finding is that there are non-linear relationships between the two dimensions of boundary-spanning search and BMI.

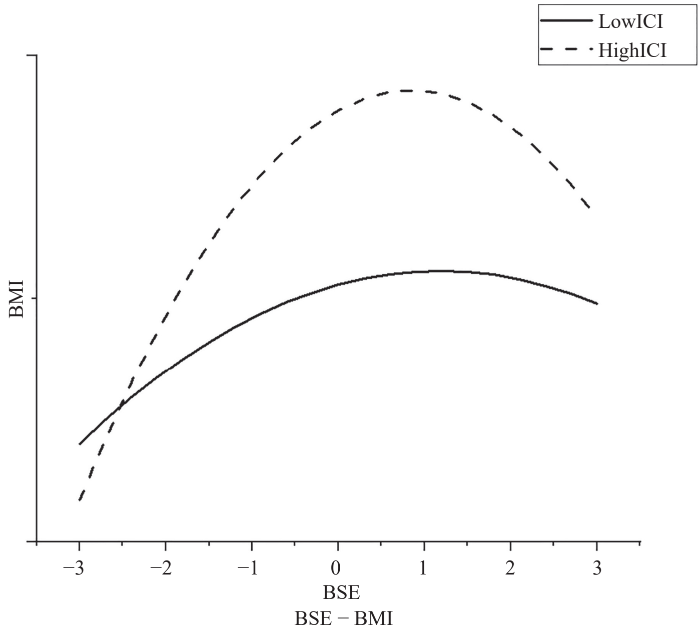
Second, ICI plays a moderating role in the relationships between BSE, BSF and BMI. Specifically, high ICI would make the inverted *U*-shaped relationships more significant. That is, higher ICI makes the enterprise more sensible about capturing BMI by searching across organizational boundaries. Meanwhile, within the moderate concentration distribution range for boundary-spanning search, the existence of high ICI would improve the level of BMI. This finding is consistent with that of Snihur and Zott (2020), who suggested that cognitive imprinting strengthens structural imprinting.

Finally, the moderating effect of ICI would be influenced by considering the context of ED. The increase in ED would weaken the moderating effect of ICI. When ED is high, the level of BMI decreases, irrespective of the ICI. This result is consistent with the finding of Sidhu *et al.* (2007) that the values of search behavior would be affected by the environmental context. Meanwhile, it is also consistent with the viewpoint in organizational imprinting research that the changing and shifting of imprints are closely related to environmental context (Marquis and Tilcsik, 2013; Simsek *et al.*, 2015; Sinha *et al.*, 2020).

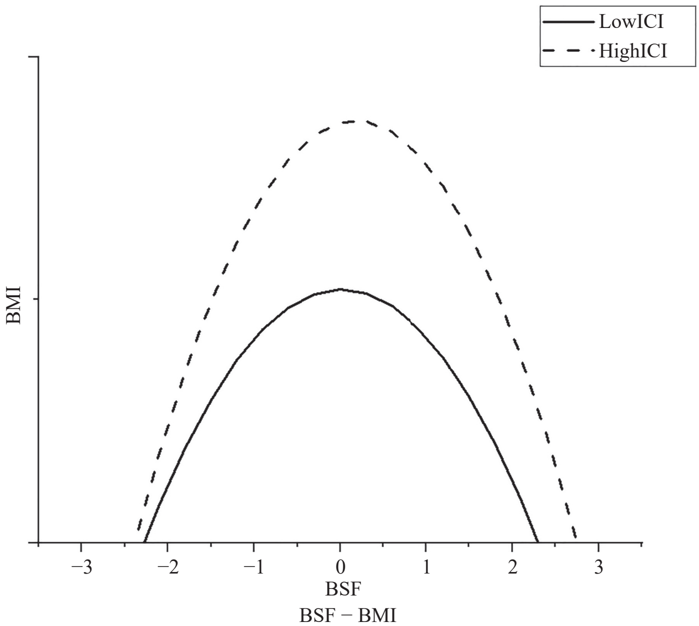
5.2 Theoretical implications

Starting from the perspective of acquiring outside exogenous knowledge, this study integrates enterprises' controlled internal active behavior factors and external occasional factors to consider the influencing mechanism of multiple factors on BMI. This study constructed a theoretical model of BSE and focus affecting BMI and introduced the joint moderating role of ICI and ED. Specifically, the findings from this study make several theoretical contributions to the current literature. First, most previous research has emphasized the influence exerted on products, procedures and other aspects of organizational search (Billinger *et al.*, 2021; Katila and Ahuja, 2002; Laursen and Salter, 2006; Li *et al.*, 2013). Few studies consider the impact of boundary-spanning search on BMI. Recent research, including Snihur and Wiklund (2019) and others, has started to focus on the role of boundary-spanning search in promoting BMI. However, the non-linear effects imposed by the varied dimensions of heterogeneous knowledge sources on organizational search have not been deeply explored. Based on organizational learning theory, this study tries to supplement existing literature by focusing on the concentration degree of knowledge sources or channels to which organizational search's dividing into different dimensions is accorded. This study tested and confirmed that both the extensity and focus of boundary-spanning search – which aims to acquire information, resources and other kinds of knowledge outside the organization boundary – have inverted *U*-shaped influences on BMI. In this way, this study proves that the extensity as well as the focus of search is important antecedents of BMI. This study is expected to be a powerful supplement to the research on the driving factors of BMI. It also corresponds to the calls for more internal antecedents of BMI made by Foss and Saebi (2017) and Foss and Saebi (2018).

Second, this study suggested and examined the moderating effects of ICI on boundary-spanning search and BMI. According to organizational imprinting theory, this study suggests and clarifies a new interpretation mechanism for the effect of cognitive imprinting. That is, ICI existing in an organization would significantly strengthen the inverted *U*-shaped relationship between the two dimensions of boundary-spanning search and BMI. While



(a)



(b)

Figure 3.
Joint moderating effect

responding to the appeal of Foss and Saebi (2017) on the moderating factors, this study extends the research of Snihur and Zott (2020) on the relationship between organizational imprinting and BMI. It also responds to research by Hsu and Lim (2014) on the joint influence of organizational knowledge and organizational imprinting on innovation. In summary, this study develops the framework that organizational learning factors influence BMI.

Another key contribution of this study lies in suggesting and verifying the joint moderating effect of ICI and ED on the relationship between boundary-spanning search and BMI. This study constructed and explained the intersection mechanism composed of organizational active learning behavior, organizational current state, passive influence of environmental change and BMI. It is revealed that high ED would weaken the moderating effect of ICI. The major insight verified the viewpoint implied in organizational imprinting theory that the changing of imprints is closely related to the organization's external environment (Boeker, 1989; Ellis *et al.*, 2017; Marquis and Tilcsik, 2013; Tilcsik, 2014). Under the framework of BMI, this is the first study to examine whether the drastic changes in the environment would have an impact on the ICI of enterprises and weaken the moderating role of ICI through empirical studies, which is hoped to provide a reference for future research.

5.3 Practical implications

Our findings also have significant practical implications for how to capture BMI. First, there is a curve relationship between BSE, BSF and BMI, which goes up first and then goes down. With the increase in the extensity and focus of boundary-spanning search, the level of BMI has also improved. Managers should encourage organizational members to acquire and absorb external knowledge, which is conducive to stimulating the generation of new ideas in the organization. Managers should also keep in mind that boundary-spanning search with a reasonable intensity is required. Literature has emphasized the importance of selective attention to external environmental stimuli (Ocasio, 2011), while the organization's energy and ability are limited. The extensity and focus of boundary-spanning search should be limited within a reasonable range; otherwise, it will exceed the organization's knowledge absorption and resource processing ability and negatively affect BMI. Therefore, strengthening the timely feedback of search results, checking the efficiency of different search methods and then reasonably allocating the attention resources of search are essential.

Second, ICI enhances the curve relationships between boundary-spanning search and BMI. The existence of ICI further improves the level of enterprise BMI. Imprinting can be re-imprinted from top to bottom for modification and coupling (De Cuyper *et al.*, 2020). Therefore, managers should pay attention to the role of ICI and strengthen the ICI of organizational members through the selection of members and construction of an innovative atmosphere, guiding members to actively search for external knowledge so as to promote the generation of new ideas and the discovery of new opportunities.

Finally, the increase in ED would weaken the moderating effect of ICI. To deal with the challenges brought by environmental change, organizations need time to absorb and precipitate. Therefore, there is a lag period between the re-imprinting of ICI and environmental change, and this has brought about challenges. Managers can use the strategic description of imprinting to adjust (including adding or modifying) organizational imprinting to adapt to the changing environment (Sinha *et al.*, 2020). Therefore, in the sensitive period when the business environment frequently changes, managers should pay more attention to communication and guidance to promote organizational members' understanding of organizational innovative objectives and vision and make existing imprints adapt to the changes in the environment.

5.4 Limitations and future research

The generalizability of these results is subject to certain limitations. The most important limitation lies in the data. The data this research used were reported by the respondents. Although common method deviation has been minimized in the process of questionnaire design and survey, there is inevitably subjectivity. Another potential issue is that the data we use are cross-sectional data, which means they cannot completely determine the causality. Therefore, we acknowledge that more objective data should be included in the research to verify the curve relationship between boundary-spanning search and BMI, and panel data could be included to judge whether there is a causality and lag effect. In addition, our data come from Chinese enterprises, which may limit the universality of our conclusions. We call for more empirical evidence to prove whether our conclusions can be extended and generalized to other economies.

There may be some intermediate mechanism between boundary-spanning search and BMI. A full discussion of antecedents, including mediators, lies beyond the scope of this study. In fact, organizational knowledge transfer (Argote *et al.*, 2021; Cohen and Levinthal, 1990) and knowledge absorptive capability (Hargadon and Sutton, 1997; Lewin *et al.*, 2020; Wu *et al.*, 2019; Zahra and George, 2002) have an impact on organizational innovation performance. They could also be included to further test the hypothesis of the inverted *U*-shape and moderating effects. Additionally, future research could also be conducted to determine the effects of enterprise decision-making methods, organizational structure characteristics and other factors on the impact of boundary-spanning search on BMI.

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